

DATABASE DESIGN ACTIVITY

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SYSTEM TITLE: LEGAL CASE MANAGEMENT SYSTEM (LCMS)

Role	Description	Key Permissions
Admin	System administrator who manages all users and has full access to the system.	Create, edit, and delete users; view all cases and reports; configure system settings
Lawyer	Registered attorney who handles assigned legal cases.	Create and manage own cases; upload documents; schedule hearings; view assigned clients.
Client	Individual seeking legal services with read-only access to their own case.	View own case status, scheduled hearings, and uploaded documents.

Guide Questions:

- Who can register?**
Lawyers and Clients register via a sign-up form. Admins are pre-seeded.
- Who can log in?**
All three roles (Admin, Lawyer, Client) can log in with email + password.
- Who manages the system?**
Admin manages user accounts and oversees all data
- Who only views information?**
Clients can only view their own case details no editing rights.
- Is there an admin or moderator?**
Yes, the Admin role has full system control including user management

5 Main Modules

- User Management Module**
Handles registration, login, and profile management for all roles. Admin can activate/deactivate accounts.
Users: Admin, Lawyer, Client
- Case Management Module**
Core module for creating, updating, and tracking legal cases. Each case has a status, assigned lawyer, and linked client.
Users: Admin, Lawyer

3. Client Management Module

Manages client records including contact info. Lawyers can add new clients. Clients can view their own profile.

Users: Admin, Lawyer, Client
4. Hearing Scheduling Module

Allows lawyers to schedule, update, and track court hearings for each case. Clients can view scheduled hearings for their case.

Users: Admin, Lawyer, Client (view)
5. Document Management Module

Upload, organize, and retrieve case-related documents (e.g., contracts, court orders). Tracks uploader and upload date.

Users: Admin, Lawyer

MAP USERS TO MODULES

Module	Admin	Lawyer	Client
User Management	✓	✓	✓
Case Management	✓	✓	✗
Client Management	✓	✓	✓
Hearing Scheduling	✓	✓	✓
Document Management	✓	✓	✗

✓ Full Access	Note: Client access to Hearing Scheduling is view-only
✗ No Access	

IDENTIFY ENTITIES (DATABASE TABLES)

Entity / Table	Description	Description
users	Stores all system users (Admin, Lawyer, Client)	user_id, full_name, email, password_hash, role, is_active, created_at
cases	Stores all legal case records.	case_id, case_number, title, description, status, filed_date, lawyer_id (FK), client_id (FK)
clients	Stores client profile information.	client_id, full_name, email, phone, address, created_at
hearings	Stores hearing schedules linked to a case.	hearing_id, case_id (FK), hearing_date, location, notes, status
documents	Stores metadata of uploaded case documents.	document_id, case_id (FK), uploaded_by (FK), file_name, file_type, uploaded_at

DEFINE BUSINESS RULES

1. Role-Based Access

Each user has a single role (admin / lawyer / client). Role is assigned at registration and can only be changed by Admin.

2. Case Ownership

A Lawyer can only view and edit cases assigned to them. Admin can view and edit all cases.

3. Client-Case Relationship

One Client can be linked to many Cases (1:M). One Case belongs to exactly one Client and one Lawyer.

4. Hearing Rules

Hearings belong to one Case. One Case can have many Hearings (1:M). Only the assigned Lawyer or Admin may create/edit hearings.

5. Document Rules

Documents are linked to a Case. Only Lawyers and Admin may upload. uploaded_by stores the user_id of the uploader.

6. User Deletion

Users cannot delete other users' records. Admin can deactivate (soft-delete) any account by setting is_active = false.

7. Admin Override

Admin can override all record restrictions: edit any case, reassign lawyers, deactivate clients.

8. Inactive Records

If a user account is set inactive (is_active = false), they cannot log in. Their existing records remain in the database.

9. Data Integrity

Deleting a Case should cascade-delete its related Hearings and Documents. Clients and Lawyers (users) are NOT deleted when a case is removed.

10. Unique Constraints

Each Case has a unique case_number. Each user must have a unique email address.

ENTITY RELATIONSHIP DIAGRAM (ERD)

